

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6492B

6 Credits: 4

Open Elective

Source-grounded

Basics of Robotics

□ Learn the types of robots, their anatomy, and key components like actuators and sensors

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 6492B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6492B.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	6492B
Course title	Basics of Robotics
Semester	6 Credits: 4
Course category	Open Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ Learn the types of robots, their anatomy, and key components like actuators and sensors
- □ Analyze factors such as axes, precision, and capacity to choose appropriate robots for various applications.
- □ Understand different sensor and actuator types and how to select them for specific robotic
- □ Learn basic kinematics principles and use transformation matrices for solving forward and inverse kinematics problems.

Course outcomes

CO	Official outcome	Study level
CO1	applications of Robots 14 Remembering Select suitable sensors and actuators for robotic	See official syllabus
CO2	14 Understanding systems. Identify the right robotic configuration and gripper	See official syllabus
CO3	14 Understanding for specific tasks. Determine the most suitable robotic configuration	See official syllabus
CO4	and gripper for a specific application. 16 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 3 3 2 2

Row	Extracted official mapping
CO3	CO3 3 3 3 2 3
CO4	CO4 3 3 3 3 3 1

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Familiarize with anatomy, specifications and applications of Robots	4	As specified
Module 2	Select suitable sensors and actuators for robotic systems.	4	As specified
Module 3	Identify the right robotic configuration and gripper for specific tasks.	4	As specified
Module 4	specific application.	5	As specified

MODULE 1

Familiarize with anatomy, specifications and applications of Robots

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize with anatomy, specifications and applications of Robots
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator,

3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering types of robots (manipulators, mobile, aerial). describe the anatomy of a robotic manipulator, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator, definition/meaning. steps, application. Technical terms English-.

▼ including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom,

including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, including links, joints, actuators, and 5 understanding controllers. explain kinematic chains, degrees of freedom, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റൂട്ട് including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom, ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം definition/meaning ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം. ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം, steps, application ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം. Technical terms English-റൂട്ട് ഡിഗ്രീസ് ഓഫ് ഫ്രീഡം.

▼ 4 Understanding and operational considerations for robots. List applications of robots in various fields like

4 Understanding and operational considerations for robots. List applications of robots in various fields like is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and operational considerations for robots. List applications of robots in various fields like using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and operational considerations for robots. list applications of robots in various fields like should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and operational considerations for robots. List applications of robots in various fields like means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding and operational considerations for robots. List applications of robots in various fields like definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.

2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining. is an official topic in Module 1 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering medical, industrial, mining, and space. definition, anatomy, and applications of robots definitions- robots, robotics; types of robots- manipulators, mobile robots-wheeled & legged robots, aerial robots; anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, repeatability, precision and accuracy, operating environment, point to point control or continuous path control., robot applications- medical, mining, space, defense, security, domestic, entertainment, industrial applications-material handling, welding, spray painting, machining. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications- Material handling, welding, Spray painting, Machining. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications- Material handling, welding, Spray painting, Machining. ഏകദേശം 15 മിനുട്ട്

definition/meaning ഉപയോഗിക്കുക. ഉപയോഗിക്കുക ഉപയോഗിക്കുക, steps, application ഉപയോഗിക്കുക ഉപയോഗിക്കുക ഉപയോഗിക്കുക. Technical terms English-ഉ ഉപയോഗിക്കുക ഉപയോഗിക്കുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Select suitable sensors and actuators for robotic systems.

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Select suitable sensors and actuators for robotic systems.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and

robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, robotics, such as touch, force, proximity, and 3 remembering vision sensors. explain the difference between internal and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-റൂട്ടിംഗ് robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-റൂട്ടിംഗ് ഏകദേശം ഏകദേശം.

▼ external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and

external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, external sensors, and describe vision sensor 4 understanding elements. describe the classification, working, and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എക്സറ്റേണൽ സെൻസറുകൾ, and describe vision sensor 4 Understanding elements. Describe the classification, working, and application of the sensor. definition/meaning എന്താണ്. പ്രവർത്തനം, steps, application എങ്ങനെ പ്രയോജനപ്പെടുത്താം. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

▼ advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and

advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, advantages/disadvantages of electric, 4 understanding hydraulic, and pneumatic actuators. apply knowledge to select suitable sensors and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്ന അനുകൂലതകളും/അനുകൂലതകളും of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and ഏതെങ്കിലും ഏതെങ്കിലും definition/meaning ഏതെങ്കിലും. ഏതെങ്കിലും ഏതെങ്കിലും ഏതെങ്കിലും, steps, application ഏതെങ്കിലും ഏതെങ്കിലും. Technical terms English-എന്ന ഏതെങ്കിലും ഏതെങ്കിലും.

▼ 3 Applying actuators for robotic applications. Sensors and Actuators
Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.

3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors- Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages. is an official topic in Module 2 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying actuators for robotic applications. sensors and actuators sensor classification- touch, force, proximity, vision sensors. internal sensors-position sensors, velocity sensors, acceleration sensors, force sensors; external sensors-contact type, noncontact type; vision - elements of vision sensor, image acquisition, image processing; selection of sensors. actuators for robots- classification-electric, hydraulic, pneumatic actuators; their advantages and disadvantages; electric actuators- stepper motors, dc motors, dc servo motors and their drivers, ac motors, linear actuators, selection of motors; hydraulic actuators- components and typical circuit, advantages and disadvantages; pneumatic actuators- components and typical circuit, advantages and disadvantages. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification- Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit,

advantages and disadvantages. ഏകദേശം definition/meaning
പ്രദേശം. ഏകദേശം പ്രദേശം, steps, application പ്രദേശം
പ്രദേശം. Technical terms English-ഏകദേശം പ്രദേശം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Identify the right robotic configuration and gripper for specific tasks.

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the right robotic configuration and gripper for specific tasks.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their

3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering (ppp, rpp, rrp, rrr) and their features. describe scara and puma robots and their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ 3 Understanding uses. Classify types of end effectors such as

3 Understanding uses. Classify types of end effectors such as is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding uses. Classify types of end effectors such as using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding uses. classify types of end effectors such as should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding uses. Classify types of end effectors such as means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding uses. Classify types of end effectors such as definition/meaning. , steps, application . Technical terms English- .

▼ 4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and

4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding grippers, tools, and their design considerations. choose appropriate configurations and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a

4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a is an official topic in Module 3 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection

and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying grippers for specific applications. robotic configurations and end effectors robot configurations-ppp, rpp, rrp, rrr; features of scara, puma robots; classification of robots based on motion control methods and drive technologies; 3r concurrent wrist; classification of end effectors - mechanical grippers, special tools, magnetic grippers, vacuum grippers, adhesive grippers, active and passive grippers, selection and design considerations of grippers in robot. determine the most suitable robotic configuration and gripper for a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations- PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a given application. **English- Malayalam** definition/meaning. **English- Malayalam** steps, application. **English- Malayalam** Technical terms. **English- Malayalam** definition/meaning. **English- Malayalam** steps, application. **English- Malayalam** Technical terms.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**specific application.**

This module is presented from the official syllabus scope for Basics of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: specific application.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using

3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3d space and their composition through 3 understanding sequential applications of rotations. explain the representation of rotations using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using definition/meaning . steps, application . Technical terms English- .

▼ 3 Understanding matrices and their properties. Understand the concept of homogeneous

3 Understanding matrices and their properties. Understand the concept of homogeneous is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding matrices and their properties. Understand the concept of homogeneous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding matrices and their properties. understand the concept of homogeneous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding matrices and their properties. Understand the concept of homogeneous means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding matrices and their properties. Understand the concept of homogeneous definition/meaning. Homogeneous definition/meaning, steps, application. Technical terms English- Malayalam.

▼ coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices

coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, coordinates and transformations for unified 4 understanding representation of positions and orientations. apply homogeneous transformation matrices should be discussed with attention to safe operation,

▼ to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to

to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, to describe translations, rotations, and scaling 4 applying in robotic systems. use forward and inverse kinematics to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിരിക്കട്ടെ, translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to എങ്ങനെയാണ് definition/meaning എങ്ങനെയാണ്. എങ്ങനെ എങ്ങനെ എങ്ങനെ, steps, application എങ്ങനെ എങ്ങനെ എങ്ങനെ. Technical terms English-എന്നിരിക്കട്ടെ എങ്ങനെ എങ്ങനെ എങ്ങനെ.

▼ **determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics - Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: SI.No Website Link <https://www.robotics.org> 1 <https://www.ros.org/> 2 <https://www.allaboutcircuits.com/textbook/direct-current/chpt-39/overview-of-sensors/> <https://www.industrial-robotics.org/> 4 [https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5 to-robotics-fall-2007/](https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5-to-robotics-fall-2007/)**

determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics -

Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No Website Link <https://www.robotics.org> 1 <https://www.ros.org/> 2 <https://www.allaboutcircuits.com/textbook/direct-current/chpt-3> 9/overview-of-sensors/ <https://www.industrial-robotics.org/> 4 <https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-to-robotics-fall-2007/> is an official topic in Module 4 of Basics of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control),

Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No

Website Link <https://www.robotics.org> 1 <https://www.ros.org/> 2

[https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/](https://www.allaboutcircuits.com/textbook/direct-current/chpt-39/overview-of-sensors/) <https://www.industrial-robotics.org/> 4

[https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/](https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5-to-robotics-fall-2007/) using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, determine the positions and orientations of 2 applying robotic arms. kinematics and motion planning understanding basic rotational motions in 2d and 3d space, composition of rotations: sequential application of rotations around different axes, representation of rotations using matrices and their properties, homogeneous coordinates and transformations, representation of positions and orientations of robotic components in a unified framework. use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. applications in robotic arm manipulation, such as forward and inverse kinematics. text /reference: t/r booktitle/author t1 s k saha ,introduction to robotics,, mc graw hill eduaction robert. j. schilling , “fundamentals of robotics – analysis and control”, t2 prentice hall of india 1996. r k mittal and i j nagrath, “robotics and control”, tata mcgraw hill, new t3 delhi, 2003. john. j. craig , introduction to robotics (mechanics and control), pearson r1 education asia 2002. ashitava ghosal, “robotics-fundamental concepts and analysis”, oxford r2 university press online resources: sl.no website link <https://www.robotics.org> 1 <https://www.ros.org/> 2 [https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/](https://www.allaboutcircuits.com/textbook/direct-current/chpt-39/overview-of-sensors/) <https://www.industrial-robotics.org/> 4 [https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/](https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5-to-robotics-fall-2007/) should be discussed with attention to safe operation, correct

terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What determine the positions and orientations of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: SI.No Website Link https://www.robotics.org 1 https://www.ros.org/ 2 https://www.allaboutcircuits.com/textbook/direct-current/chpt-3/overview-of-sensors/ https://www.industrial-robotics.org/ 4 https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5 to-robotics-fall-2007/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 of 2 Applying robotic arms. Kinematics and Motion Planning Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics. Text /Reference: T/R BookTitle/Author T1 S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction Robert. J. Schilling , “Fundamentals of robotics – Analysis and control”, T2 Prentice Hall of India 1996. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New T3 Delhi, 2003. John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson R1 Education Asia 2002. Ashitava Ghosal, “Robotics-Fundamental concepts and analysis”, Oxford R2 University press Online resources: Sl.No Website Link <https://www.robotics.org> 1 <https://www.ros.org/> 2 [https://www.allaboutcircuits.com/textbook/direct-current/chpt- 3 9/overview-of-sensors/](https://www.allaboutcircuits.com/textbook/direct-current/chpt-39/overview-of-sensors/) <https://www.industrial-robotics.org/> 4 [https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- 5 to-robotics-fall-2007/](https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-5-to-robotics-fall-2007/) [റോബോട്ടിക്സ് ഫണ്ടമെന്റൽ കോൺസെപ്റ്റ്സ് definition/meaning](#) [റോബോട്ടിക്സ് ഫണ്ടമെന്റൽ കോൺസെപ്റ്റ്സ്](#). [റോബോട്ടിക്സ് ഫണ്ടമെന്റൽ കോൺസെപ്റ്റ്സ്](#), steps, application [റോബോട്ടിക്സ് ഫണ്ടമെന്റൽ കോൺസെപ്റ്റ്സ്](#) [റോബോട്ടിക്സ്](#). Technical terms English-[റോബോട്ടിക്സ് ഫണ്ടമെന്റൽ കോൺസെപ്റ്റ്സ്](#).

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator,. (3 marks)**

► **Define or explain including links, joints, actuators, and 5 Understanding controllers. Explain kinematic chains, degrees of freedom,. (3 marks)**

► **Define or explain 4 Understanding and operational considerations for robots. List applications of robots in various fields like. (3 marks)**

► **Define or explain 2 Remembering medical, industrial, mining, and space. Definition, Anatomy, and Applications of Robots Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots- wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.. (3 marks)**

Module 2

► **Define or explain robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and. (3 marks)**

► **Define or explain external sensors, and describe vision sensor 4 Understanding elements. Describe the classification, working, and. (3 marks)**

► **Define or explain advantages/disadvantages of electric, 4 Understanding hydraulic, and pneumatic actuators. Apply knowledge to select suitable sensors and. (3 marks)**

► **Define or explain 3 Applying actuators for robotic applications. Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors. Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.. (3 marks)**

Module 3

► **Define or explain 3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their. (3 marks)**

► **Define or explain 3 Understanding uses. Classify types of end effectors such as. (3 marks)**

► **Define or explain 4 Understanding grippers, tools, and their design considerations. Choose appropriate configurations and. (3 marks)**

► **Define or explain 4 Applying grippers for specific applications. Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot. Determine the most suitable robotic configuration and gripper for a. (3 marks)**

Module 4

► **Define or explain 3D space and their composition through 3 Understanding sequential applications of rotations. Explain the representation of rotations using. (3 marks)**

► **Define or explain 3 Understanding matrices and their properties. Understand the concept of homogeneous. (3 marks)**

► **Define or explain coordinates and transformations for unified 4 Understanding representation of positions and orientations. Apply homogeneous transformation matrices. (3 marks)**

► **Define or explain to describe translations, rotations, and scaling 4 Applying in robotic systems. Use forward and inverse kinematics to. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Remembering types of robots (manipulators, mobile, aerial). Describe the anatomy of a robotic manipulator,**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **robotics, such as touch, force, proximity, and 3 Remembering vision sensors. Explain the difference between internal and.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Remembering (PPP, RPP, RRP, RRR) and their features. Describe SCARA and PUMA robots and their.**

Module 4

1-mark definition: State the meaning of **3D space and their composition through 3 Understanding sequential applications of rotations. Explain**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

the representation of rotations using.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.robotics.org> <https://www.ros.org/>
- <https://www.allaboutcircuits.com/textbook/direct-current/chpt- /overview-of-sensors/> <https://www.industrial-robotics.org/>
- <https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction- to-robotics-fall-2007/>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 6492B. Verify institutional updates against the current official curriculum.